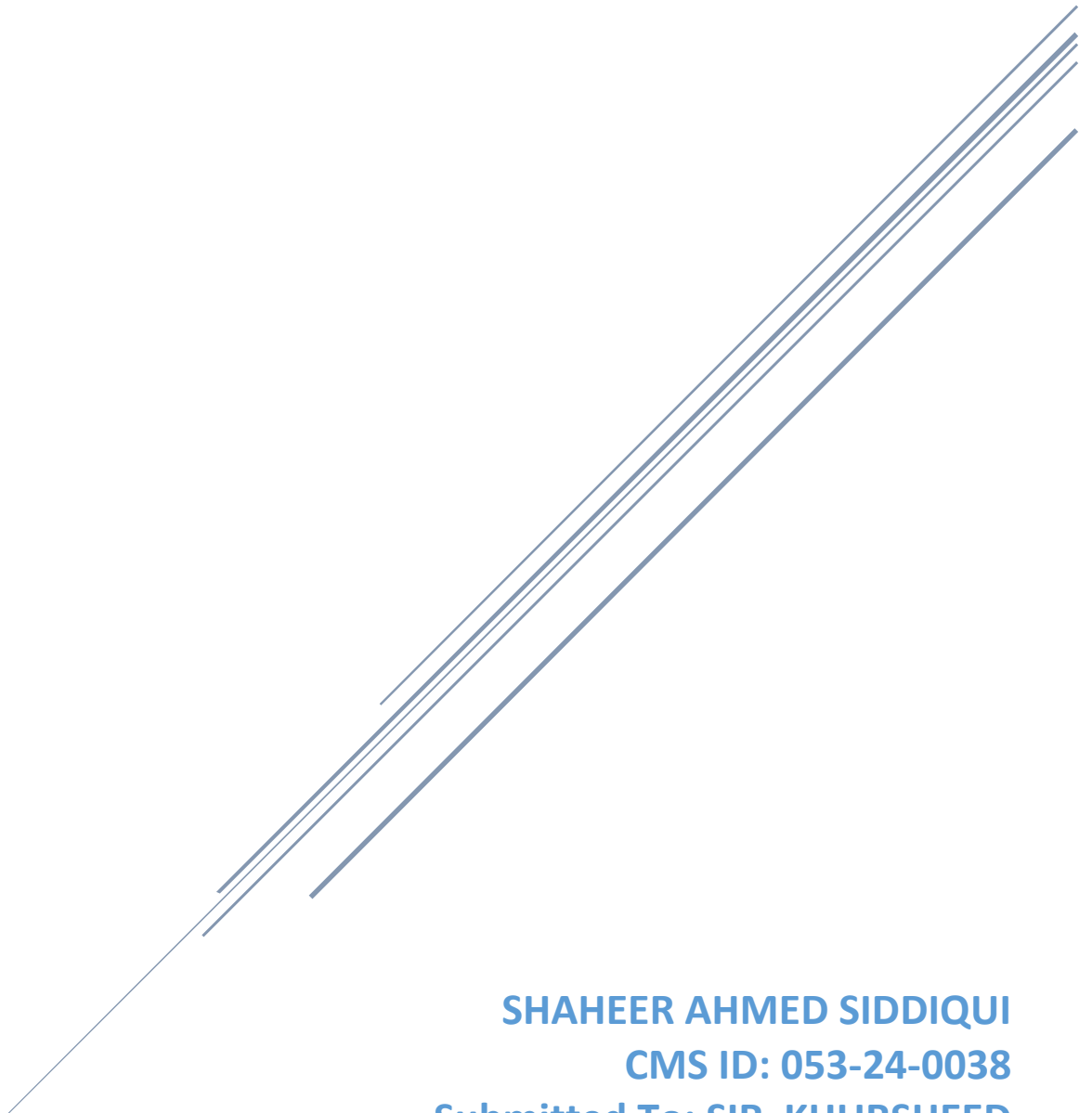


SUKKUR IBA UNIVERSITY

SOFTWARE QUALITY ENGINEERING

LAB-1



SHAHEER AHMED SIDDIQUI

CMS ID: 053-24-0038

Submitted To: SIR KHURSHEED

Table of Contents

Exercise of SQE Lab 05.....	2
GitHub Repository URL:	2
Repository Structure:	2
Overall Structure:	2
Local Repository Structure:	3
GitHub Repository Structure:	4
Three or more issues:.....	4
GitHub Project Board:	5
README file:.....	5
Git Command Execution Evidence:.....	8
Questions & Answers:.....	10
1. What is Git?	10
2. What is GitHub?	10
3. What is the difference between Git and GitHub?	10
4. Why should .gitignore be used?	10
5. What is an issue in GitHub?	10
6. Why are labels useful for SQE?	11

Exercise of SQE Lab 05

GitHub Repository URL:

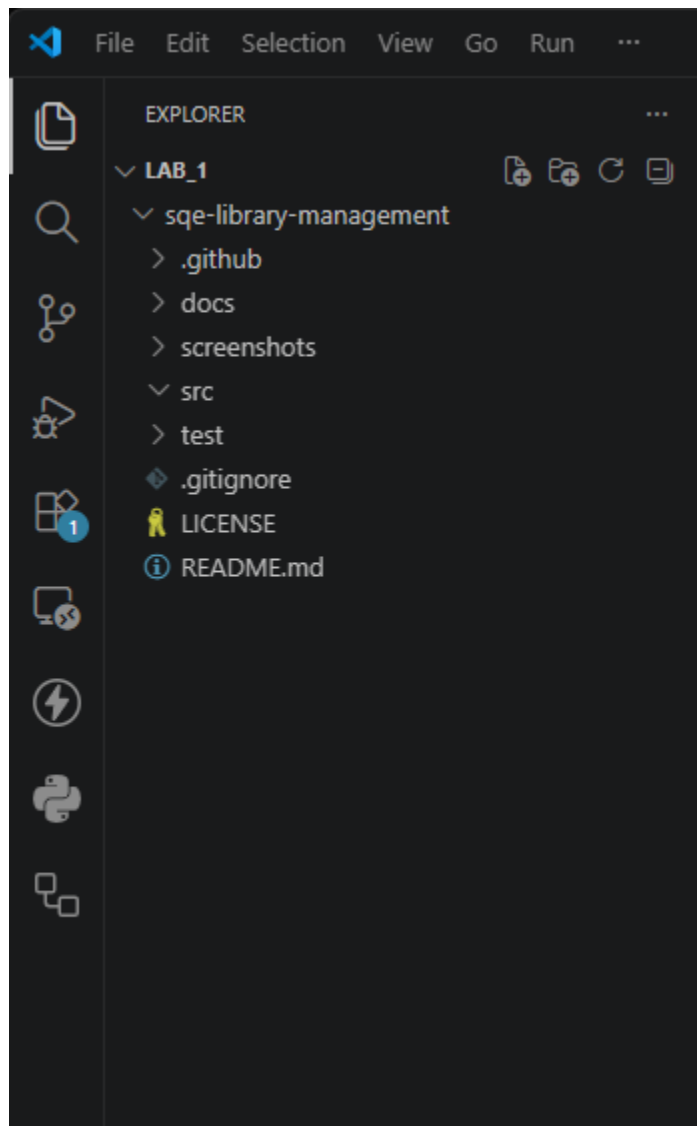
<https://github.com/ShahmeerAhmedSiddiqui/sqe-library-management>

Repository Structure:

Overall Structure:

```
sqe-library-management/
|
|— .github/
|   └─ workflows/
|
|— docs/
|   └─ SQE_Lab_01_Report.docx
|
|— screenshots/
|   └─ commits.png
|   └─ Issues.png
|   └─ Labels.png
|   └─ Project_board.png
|   └─ Project_Structure.png
|   └─ Repository.png
|
|— src/
|— tests/
|— .gitignore
|— LICENSE
└─ README.md
```

Local Repository Structure:



GitHub Repository Structure:

main	1 Branch	0 Tags	Go to file	Add file	Code
ShaheerAhmedSiddiqui add screenshot and extras 7eb2b9f · 12 minutes ago 3 Commits					
docs	add screenshot and extras		12 minutes ago		
screenshots	add screenshot and extras		12 minutes ago		
src	add screenshot and extras		12 minutes ago		
.gitignore	Initial commit		1 hour ago		
LICENSE	Initial commit		1 hour ago		
README.md	Add ReadMe.ms		14 minutes ago		

Three or more issues:

ShaheerAhmedSiddiqui / sqe-library-management

Code Issues 2 Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Issues

Assigned to me

Created by me

Mentioned

Recent activity

Views

Projects

Milestones

Labels

All issues

is:issue

Open 2 Closed 1

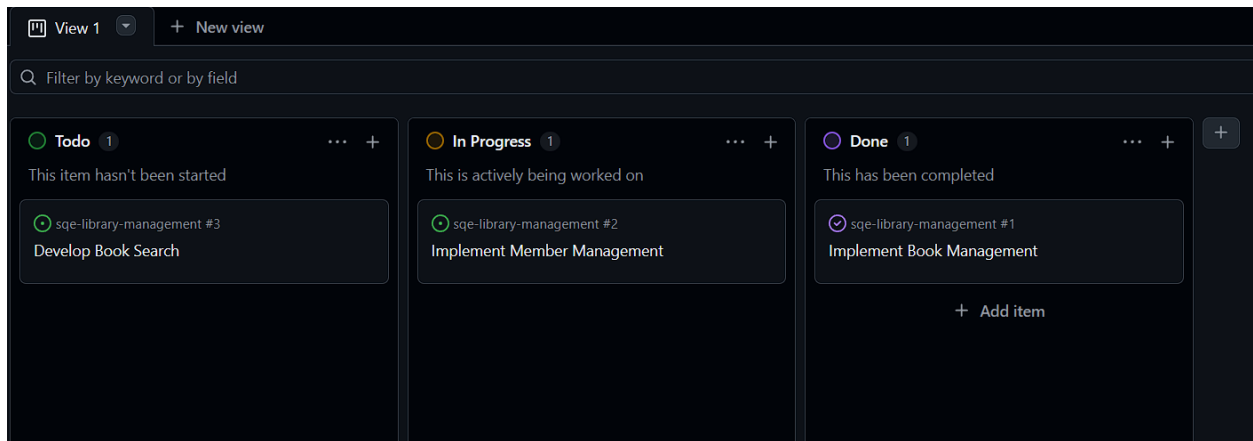
Author Labels Projects Milestones Assignees Newest

☐ Develop Book Search accessibility enhancement High Priority #3 · ShaheerAhmedSiddiqui opened 1m ago

☐ Implement Member Management enhancement High Priority #2 · ShaheerAhmedSiddiqui opened 1m ago

☒ Implement Book Management enhancement #1 · by ShaheerAhmedSiddiqui was closed 2m ago

GitHub Project Board:



README file:

SQE Library Management

Project Description

The **SQE Library Management System** is a software project developed for the **Software Quality Engineering (SQE)** course. The project is focused on understanding and applying software quality practices through a structured **GitHub-based development workflow**.

The system is designed to support common library operations such as managing books, managing library members, and searching for available books.

Main Features

- **Book Management** — Add, update, delete, and view book records.
- **Member Management** — Add, update, delete, and view member information.
- **Book Search** — Search for books and check their availability.
- **Issue Tracking** — Use GitHub Issues to track development tasks and problems.
- **Quality Management** — Apply testing, documentation, and software quality practices throughout the project.

Project Structure

sqe-library-management/

|

├── src/

```
|   └─ Application source code
|
|   └─ tests/
|       └─ Test cases and automated tests
|
|   └─ docs/
|       └─ Project documentation
|
|   └─ screenshots/
|       └─ Screenshots and evidence of project activities
|
|   └─ .github/
|       └─ workflows/
|           └─ GitHub Actions workflows
|
|   └─ README.md
|   └─ LICENSE
└─ .gitignore
```

Directory Purpose

Directory/File	Purpose
src/	Contains the application source code
tests/	Contains test cases and automated tests
docs/	Contains project and quality-related documentation
screenshots/	Stores screenshots and evidence of project activities
.github/workflows/	Contains GitHub Actions workflow files

Directory/File	Purpose
README.md	Provides project information and documentation
LICENSE	Defines the project's licensing terms
.gitignore	Specifies files and folders that Git should not track

GitHub Project Management

GitHub is used to manage the development and quality activities of this project.

The repository uses:

- **GitHub Issues** for tracking development tasks
- **Labels** for categorizing issues
- **GitHub Projects** for monitoring task progress
- **Git** for version control
- **GitHub Actions** for future automation and continuous integration

Initial Issues

The following issues are included in the project:

1. **Implement Book Management**
2. **Implement Member Management**
3. **Develop Book Search**

Project Labels

The project uses the following labels:

- bug
- enhancement
- testing
- documentation
- high-priority
- quality

Project Objective

The main objective of this project is to practice **Software Quality Engineering** concepts within a professional software development workflow.

The project focuses on:

- Version control using Git and GitHub
- Issue and task management
- Software testing
- Quality assurance activities
- Project documentation
- Continuous integration
- Collaborative software development
- Tracking and improving software quality

Technologies & Tools

- **Git**
- **GitHub**
- **GitHub Issues**
- **GitHub Projects**
- **GitHub Actions**
- **Markdown**

Course Information

Course: Software Quality Engineering (Lab)

Lab: Lab 01 — GitHub Repository and SQE Project Setup

Author

Shaheer Ahmed Siddiqui

BS Software Engineering

Sukkur IBA University

Git Command Execution Evidence:

The required Git tasks were successfully performed during the lab. The repository was first cloned from GitHub, after which the project directory was accessed and the repository was initialized. The project

was then synchronized with the remote main branch using `git pull origin main`. The repository status was checked using `git status`, followed by reviewing changes with `git diff`. The newly created project files and folders were staged using `git add .` and committed with the message **"add screenshot and extras"** using `git commit`. Finally, the changes were successfully uploaded to the GitHub repository using `git push origin main`. The screenshots below provide evidence of these Git operations and their corresponding results.

```
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git clone https://github.com/ShaaheenAhmedSiddiqui/sqe-library-management.git
Cloning into 'sqe-library-management'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git pull origin main
fatal: not a git repository (or any of the parent directories): .git
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git inti
git: 'inti' is not a git command. See 'git --help'.

The most similar command is
    inti
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git init
Initialized empty Git repository in E:/universityMaterial/5thSemester/SoftwareQualityEngineering/SQE_Labs/lab_1/.git/
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git pull origin
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> cd sqe-library-management
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> cd sqe-library-management
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git init
Reinitialized existing Git repository in E:/universityMaterial/5thSemester/SoftwareQualityEngineering/SQE_Labs/lab_1/sqe-library-management/.git/
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 2.23 KiB | 14.00 KiB/s, done.
From https://github.com/ShaaheenAhmedSiddiqui/sqe-library-management
* branch      main      -> FETCH_HEAD
   cd0ce7e..ff8df10  main  -> origin/main
Updating cd0ce7e..ff8df10
Fast-forward
```

```
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> cd sqe-library-management
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git pull origin main
README.md | 124 ++++++
1 file changed, 122 insertions(+), 2 deletions(-)
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    docs/
    screenshots/
    src/

nothing added to commit but untracked files present (use "git add" to track)
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git diff
git: 'diff' is not a git command. See 'git --help'.

The most similar command is
    inti
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git diff
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git add .
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git commit -m "add screenshot and extras"
[main 7eb2b9f] add screenshot and extras
8 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 docs/ShaaheenAhmed_SQE_LAB-1_Report_053-24-0038.docx
create mode 100644 docs/~ShaaheenAhmed_SQE_LAB-1_Report_053-24-0038.docx
create mode 100644 screenshots/Project_board.png
create mode 100644 screenshots/initialize.png
create mode 100644 screenshots/issues.png
create mode 100644 screenshots/labels.png
create mode 100644 screenshots/projectStructure.png
create mode 100644 src/index.js
PS E:\universityMaterial\5thSemester\SoftwareQualityEngineering\SQE_Labs\lab_1> git push origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 4 threads
```

```
PS E:\Semester_05\Software Quality Engineering\Labs\Lab_01> cd sqe-library-management
PS E:\Semester_05\Software Quality Engineering\Labs\Lab_01> git commit -m "Readme file is updated"
[main a317da5] Readme file is updated
4 files changed, 47 insertions(+), 1 deletion(-)
rename {screenshots => screenshots}/loginpage.jpg (100%)
delete mode 100644 test/test.java
create mode 100644 tests/test.java
```

Questions & Answers:

1. What is Git?

Git is a distributed version control system used to track changes in source code and other project files. It allows developers to maintain different versions of their work, create branches, make commits, and restore previous versions when required.

2. What is GitHub?

GitHub is an online platform for hosting Git repositories. It provides developers with tools to store projects, collaborate with others, manage Issues and Projects, review code, and track the progress of software development.

3. What is the difference between Git and GitHub?

Git and GitHub are closely related, but they serve different purposes. **Git** is a version control system that runs primarily on a developer's local computer and is used to track and manage changes in a project. **GitHub** is a cloud-based platform that hosts Git repositories and provides additional collaboration and project management features.

For example, Git is used to create commits, manage branches, and track changes locally, while GitHub is used to store the repository online, create Issues, manage Projects, and collaborate with other developers.

4. Why should .gitignore be used?

The .gitignore file tells Git which files and directories should **not be tracked or committed** to the repository. It helps keep the repository organized by excluding unnecessary or automatically generated files, such as temporary files, build files, IDE-specific files, and sensitive configuration files.

For example, compiled files, build directories, and environment configuration files can be added to .gitignore so that they are not accidentally committed to GitHub.

5. What is an issue in GitHub?

A **GitHub Issue** is a feature used to create, track, and manage tasks, bugs, feature requests, improvements, and other project-related work. Issues help the development team organize responsibilities and monitor the progress of different activities.

In this project, Issues were created for **Implement Book Management**, **Implement Member Management**, and **Develop Book Search**. The **Implement Book Management** issue was assigned the **high-priority** label because it should be addressed before the other tasks.

6. Why are labels useful for SQE?

Labels are useful in **Software Quality Engineering (SQE)** because they help categorize and organize GitHub Issues based on their type, priority, or purpose. This makes it easier to identify important tasks, track quality-related activities, and manage project work efficiently.

In this project, labels such as **bug**, **enhancement**, **testing**, **documentation**, **high-priority**, and **quality** are used to classify Issues and make project management more organized.

THE END